

From Tool Invocation to Source-Mechanism Exploration: Protected White-Box DSE for Open-Source EDA

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Abstract

Open-source EDA tools allow design-space exploration (DSE) to move beyond public knobs and into bounded source-level mechanisms inside staged optimizers. We present ReviewDSE, a protected white-box DSE framework that explores such mechanisms for a target design. ReviewDSE evaluates complete source candidates under a protected evaluator and records reusable search knowledge as reviewed mechanism-level evidence. It first constructs method evidence and source-start branches from calibration designs, then uses these fixed warm-start products to initialize target-case exploration under Teacher review and full-flow validation. We instantiate ReviewDSE on OpenROAD detailed placement as a representative staged open-source EDA optimizer. Across nine target tasks, ReviewDSE reduces final post-DPL half-perimeter wirelength (HPWL) by 1.78% on average under a 2× runtime gate, compared with 0.38% for public-knob black-box DSE. A runtime-aware ReviewDSE selection retains a 1.68% reduction at 1.11× runtime, and full-flow review exposes stage-composability failures while source-mechanism exploration repairs hard cut-row legality failures.

1 Introduction

Design-space exploration (DSE) is a standard methodology for improving electronic-design-automation (EDA) flows [1]. Conventional DSE treats each tool as a fixed executable and searches its public control surface, such as command options, flow scripts, or other exposed knobs, to find configurations with better quality of results [2]. This black-box formulation is practical and broadly applicable, especially for closed-source tools, but it leaves quality-critical implementation choices inside large EDA codebases outside the search space.

Open-source EDA changes where the DSE boundary can be drawn. In an open codebase such as OpenROAD [3], an optimizer is not only invocable but also inspectable and modifiable. This enables white-box DSE for open-source EDA: instead of searching only over public knobs of a fixed executable, the search can enter selected parts of the optimizer implementation and explore bounded source-level mechanisms. In EDA flows, this mechanism space is naturally organized by stages and stage interfaces. A placement flow, for example, may include legalization policies, local improvement kernels, objective scoring, displacement control, rollback logic, and state passed from legalization to detailed-placement optimization. White-box DSE therefore asks which internal mechanisms should be changed, where they should be inserted, and how they should be configured for a target design.

The expanded mechanism space is difficult to explore manually. Source-level choices are numerous, implementation-dependent, and strongly coupled across stages. A local improvement in one stage

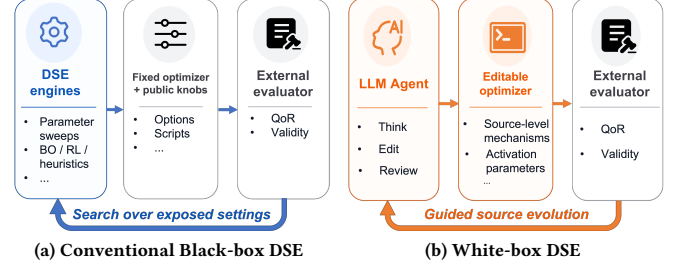


Figure 1: Search boundary for one target: black-box DSE tunes public knobs; white-box DSE explores bounded internal mechanisms under the protected evaluator.

may not survive the downstream flow; for example, a legalizer that reduces post-legalization HPWL can still produce a placement that is harder for detailed-placement optimization (DPO) to improve. Recent coding-evolution systems suggest a possible path forward: language-model agents can modify programs using evaluator feedback [4–6], and related ideas have begun to appear in EDA source-level evolution [7–10]. These systems show that agents can generate useful code changes for complex tools.

Source modification alone does not make a reliable DSE procedure. Existing loops often treat each generated diff as a complete variant to be selected or rejected as a whole. This whole-variant view gives limited reusable evidence about which mechanism caused a gain, which stage activated it, whether it was live, whether it was runtime-expensive, or why it failed to compose with downstream optimization. Unconstrained source editing can also corrupt the evaluator path, change scripts or parsers, bypass legality checks, or produce stale source/binary/metric mismatches. Reliable white-box DSE therefore needs both a protected evaluator and a review unit finer than the complete diff.

We present ReviewDSE, an agentic white-box DSE framework for open-source EDA. ReviewDSE evaluates complete source candidates under the protected evaluator, while recording reusable search knowledge as reviewed stage/interface mechanism evidence. This evidence captures source references, liveness signals, compatibility observations, and promotion or rejection decisions. ReviewDSE uses a Teacher–Student agent structure: the Teacher reviews full-flow behavior and mechanism evidence, while Student agents adapt source starts, borrow peer mechanisms, or repair failed edits.

ReviewDSE follows a two-level workflow. Level 1 builds method evidence and source-start branches from calibration designs. Level 2 conducts target-design exploration through initialization based on the frozen evidence from Level 1, followed by iterative search driven by target-specific metrics, logs, diffs, peer candidates, and Teacher review. We instantiate ReviewDSE on OpenROAD detailed placement,

where legalization, DPO, and their handoff expose strict legality and stage-composability trade-offs.

This paper makes the following major contributions:

- We formulate protected white-box DSE for open-source EDA, where the search space extends from public tool controls to bounded source-level mechanisms inside staged optimizers.
- We introduce ReviewDSE, a two-level source-mechanism exploration framework that builds calibration-time method evidence and source-start branches, then uses them to initialize target-specific exploration with Teacher review, liveness checks, full-flow validation, and target-local evidence.
- We instantiate ReviewDSE on OpenROAD detailed placement and show that reviewed source-mechanism exploration improves the post-DPL HPWL/runtime frontier beyond public-knob BO-DSE, exposes why stage-local winners require full-flow review, and recovers legality on hard cut-row stress tests.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 presents ReviewDSE. Section 4 reports the results. Sections 5 and 6 discuss limitations and conclude the paper.

2 Background and Related Work

2.1 DSE in EDA

Design space exploration (DSE) is widely used in EDA because QoR is highly sensitive to algorithm choices, tool parameters, and flow-stage interactions [1]. Prior work has explored DSE at multiple abstraction levels. In high-level synthesis, AutoDSE [11] searches HLS pragmas and transformations through bottleneck-guided exploration. In logic synthesis, BOiLS [12] formulates synthesis-flow selection as a sequential optimization problem and searches logic-optimization recipes with Bayesian optimization. In physical design, placement-tuning methods optimize placement parameters and objectives using learning-based or multi-objective search [13, 14]. These works show that automated search can improve QoR and reduce manual effort across EDA flows.

However, most existing EDA DSE methods remain black-box with respect to the optimizer implementation. Their search spaces are defined by exposed pragmas, parameters, command sequences, scripts, seeds, or flow-level configurations, while the tool source remains fixed. Such methods can improve how tools are invoked, but cannot directly explore internal mechanisms such as hidden cost models, repair policies, rollback decisions, or stage interfaces. Open-source infrastructures such as OpenROAD [3] make these implementation-level decisions searchable under the same external flow metrics, motivating white-box DSE where bounded source-level mechanisms become design choices.

2.2 LLM Agents for EDA

Large language models have recently been explored in EDA as natural-language interfaces, code generators, and tool-assistance agents. One major line of work studies RTL or Verilog generation from natural-language specifications, with benchmarks such as RTLLM [15] and VerilogEval [16] evaluating syntax, functional correctness, and design quality, and domain-specific models such as VeriGen [17] and RTLCoder [18] improving HDL generation through EDA-specific training. Beyond direct code generation, feedback-driven frameworks use compiler or simulation outputs to repair

Table 1: Positioning of ReviewDSE.

Category	Search point	Review signal	Search memory
EDA parameter DSE	knobs/scripts	QoR logs	best settings
LLM tool use	artifacts/scripts	tool logs	revised artifact
Source evolution	source patch	build/QoR logs	winning patch
ReviewDSE	source mechanism edit	protected full-flow evidence	reusable mechanism evidence

generated HDL, as in AutoChip [19]. LLMs have also been used for hardware verification and testbench generation [20, 21]. Another line of work uses LLMs as EDA tool assistants: ChatEDA [22] supports task decomposition, script generation, and tool execution, ChipNeMo [23] studies domain-adapted assistants for chip-design tasks, and documentation-grounded systems answer questions over EDA manuals and OpenROAD documentation [24, 25]. These works improve how designers generate artifacts, query documentation, and invoke EDA tools, but they mostly operate around a fixed executable.

Recent coding-evolution systems show that LLM agents can modify programs using evaluator feedback [4–6]. Similar ideas have begun to appear in EDA source-level evolution [7–10]. In this setting, the LLM action moves from scripts or artifacts around the tool to the implementation of the tool itself. Self-Evolved ABC [9] evolves a logic-synthesis codebase under correctness and synthesis-QoR feedback. AuDoPEDA [10] produces repository-grounded OpenROAD diffs using code-graph documentation and QoR-driven execution. GR-Evolve [8] specializes global-router source code to individual designs, and EvoPlace [7] evolves global-placement optimization components.

These works establish that LLM agents can improve EDA tools through source modification. ReviewDSE addresses a complementary question: how to make source modification a reliable white-box DSE procedure. Rather than treating each generated diff only as a whole-program winner or loser, ReviewDSE evaluates complete candidates under a protected flow while recording reusable positive and negative evidence at the mechanism level. This allows useful calibration evidence, stage donors, and rejection decisions to guide later search rounds. Table 1 summarizes the position of ReviewDSE relative to prior directions. ReviewDSE still evaluates complete source candidates, but its reusable memory is mechanism-level evidence that later agents can refine or combine during white-box optimization.

3 Methodology

3.1 Formulation and Overview

Conventional black-box DSE treats the EDA optimizer as a fixed executable evaluated by an external flow-level evaluator. Let O denote the implementation of a staged optimizer for a target design d , let E denote the protected evaluator used to run the flow and collect canonical metrics, and let $x \in \mathcal{X}$ denote the public configuration vector exposed through command options, flow scripts, pass sequences, or other tool parameters. Black-box DSE searches over $x \in \mathcal{X}$. Each search point is evaluated as $E(O, x, d)$, while the optimizer implementation O and the measurement path E remain fixed. This formulation is practical, but it limits the search to behaviors already exposed by the tool developer.

ReviewDSE keeps the same evaluator E but expands the search variable from a public configuration to a bounded source edit. Let $\mathcal{S}$ denote the editable optimizer source surface and let $\mathcal{D}(\mathcal{S})$ denote the

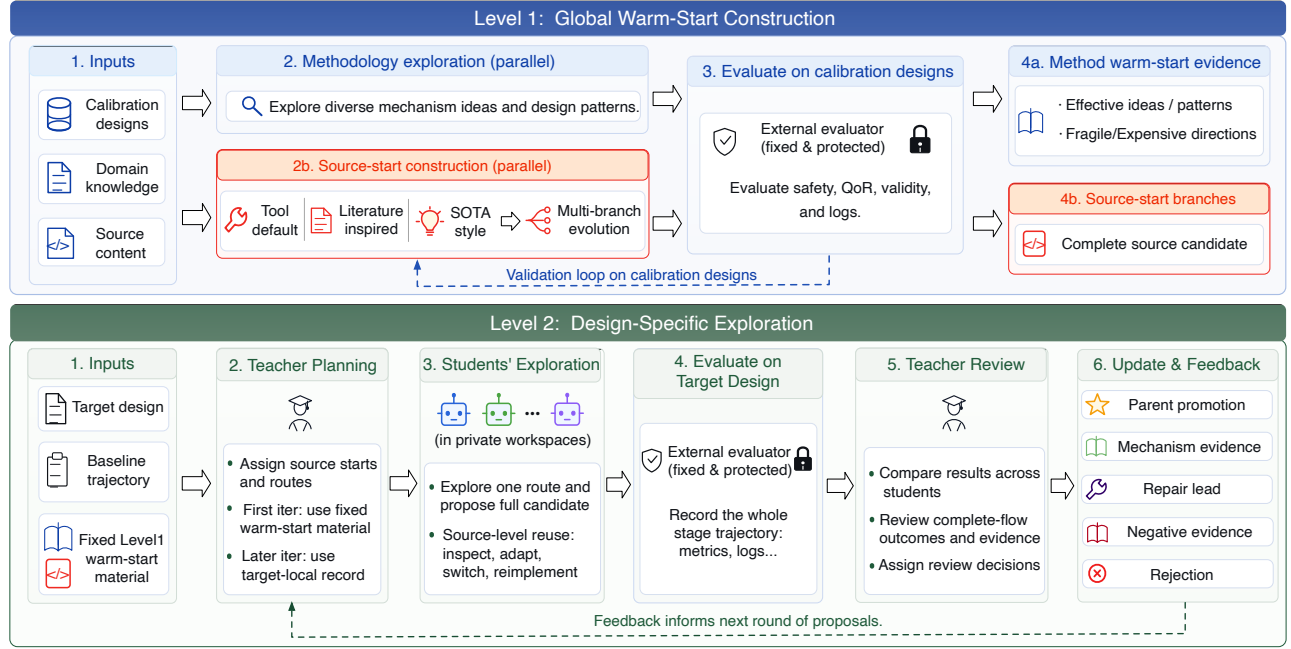


Figure 2: ReviewDSE workflow. Level 1 builds fixed warm-start evidence and source-start branches. Level 2 initializes target search from them, then explores candidates under the protected evaluator.

admissible edits on that surface. ReviewDSE searches over $\Delta \in \mathcal{D}(\mathcal{S})$. Each search point produces a complete source candidate $O + \Delta$, which is built and evaluated on d as $E(O + \Delta, d)$.

EDA optimizers are naturally organized as staged flows, with intermediate states passed across neighboring stages. ReviewDSE uses this native structure to define review scopes for source edits. Let $\mathcal{C}$ denote the set of stage and interface scopes, where each $c \in \mathcal{C}$ corresponds to either an existing optimization stage or a handoff between two neighboring stages. Each scope has a pool of candidate mechanisms,

$$\mathcal{M}_c = \{p_{c,1}, \dots, p_{c,n_c}\}. \quad (1)$$

A mechanism is not the stage itself. It is a source-level logic chain that changes how the stage initializes state, generates candidates, scores moves, accepts or rolls back transactions, exports handoff state, or performs local repair. A mechanism may also be controlled by activation parameters $\theta_{c,j}$, such as weights, thresholds, budgets, seeds, fallback rules, or early-stop conditions.

A candidate edit Δ may touch one or more native scopes. For each scope c , let $\mathcal{M}_c(\Delta) \subseteq \mathcal{M}_c$ be the mechanisms implemented by the source candidate produced by Δ . The mechanism content of the candidate can be written as

$$\text{mech}(\Delta) = \bigcup_{c \in \mathcal{C}} \{(c, p_{c,j}, \theta_{c,j}) \mid p_{c,j} \in \mathcal{M}_c(\Delta)\}. \quad (2)$$

This notation describes the mechanisms realized by an implemented source candidate. It does not imply that ReviewDSE materializes standalone mechanism units and assembles them into a tool variant. The actual search point is a source workspace built and evaluated as $O + \Delta$. After evaluation, ReviewDSE retains mechanism-level evidence rather than reusable code objects. As summarized in Table 2, this evidence records source anchors, liveness signals, full-flow behavior, and review outcomes, allowing later search to inspect, adapt,

Table 2: Reviewed mechanism evidence record.

Field	Recorded information
Scope	Target stage or stage interface, used to relate evidence to stage-wise behavior.
Source reference	Branch or patch reference, modified files, code anchors, and mechanism intent.
Controls/liveness	Activation parameters, counters, traces, and artifacts showing execution.
Review outcome	Full-flow metrics, compatibility observations, and Teacher decisions such as parent promotion, mechanism evidence, repair lead, negative evidence, or rejection.

or reimplement the reviewed mechanism logic in another candidate. A record may indicate that the mechanism improves the target, is active but harmful, never executes, depends on a compatible handoff, or should be avoided.

3.2 Protected Evaluation Contract

The protected evaluator is the trust boundary for source-level DSE. It fixes the benchmark inputs, incoming design state, top-level command sequence, baseline harness, legality or validity checker, metric parser, reference comparison, runtime gate, evaluator scripts, and unrelated tool components. Evaluator-produced objective and stage metrics remain outside the editable source surface. A candidate is rejected if it changes or bypasses this path, fails to build, violates validity checks, misses canonical metrics, exceeds the runtime gate, or lacks liveness evidence for its claimed mechanism.

Figure 2 summarizes how ReviewDSE works. Level 1 constructs frozen warm-start evidence before target optimization: method evidence records calibration-observed mechanism routes, and source-start construction produces complete source basins. Level 2 uses this evidence as prior guidance, but every promoted candidate is rebuilt, executed, and reviewed on the current target case. Level 2

Table 3: Teacher review decisions used in exploration.

Decision	Condition	Use
Parent promotion	Valid, complete metrics, live, within gate, im-	Next parent
Mechanism evidence	proves target score Active mechanism helps a stage or interface but loses final score	Evidence record
Repair lead	Explains validity/runtime failure or fragile hand-off	Targeted route
Negative evidence	Inactive, too expensive, or non-composable	Search constraint
Candidate rejection	Build, validity, metric, stale-binary, or evaluator-path failure	Discard

may accumulate target-local evidence, but it does not update the global warm-start evidence.

3.3 Calibration Warm-Start Construction

Level 1 prepares the search by building two fixed warm-start products before any target case is optimized: method warm-start and source-start branches. It does so by using calibration designs as diagnostic probes. These designs are not target designs. Instead, they are constructed to expose different optimizer behaviors, including stage-local damage, downstream recoverability, constraint pressure, runtime pressure, and stage-interface incompatibility. Level 1 converts these observations into mechanism hypotheses and executable starting points, so that Level 2 does not begin from a blank prompt.

Method warm-start construction focuses on the hypothesis side. Agents inspect papers, source code, and domain knowledge, then implement candidate mechanisms on existing branches and evaluate them with the protected flow. Level 1 review summarizes the calibration behavior of each idea, including whether it is promising, fragile, too expensive, inactive, or non-composable. The resulting records provide route and diagnosis evidence for the Level 2 Teacher when a target case shows similar stage movement or design characteristics.

Source-start construction focuses on the implementation side. Some literature-inspired or SOTA-style ideas are too large to recreate in each target iteration, so Level 1 materializes them as complete source branches during calibration. In Level 2, these branches serve as editable starts rather than certified target solutions. Students may extend one, switch to another, or reimplement a useful logic chain after inspecting peer diffs. This gives the target search concrete code for implementing or repairing a route suggested by the method evidence.

3.4 Target-Case Source Exploration Loop

Level 2 runs an independent white-box optimization loop for each target design. For a new target, ReviewDSE first measures the target baseline trajectory and creates private Student workspaces. The loop maintains a local record that includes baseline metrics, evaluated candidates, stable source references, implementation diffs, Student knowledge cards, Teacher reviews, peer-learning packets, and any clean promoted parent. The Level 1 warm-start material is only used as prior knowledge, and subsequent decisions are driven by the local record accumulated on the current design.

In the first iteration, the Teacher compares the target baseline trajectory and design characteristics with the fixed Level 1 method evidence and available source branches. Based on this comparison, the Teacher assigns each Student an initial source start and mechanism route, with generated packets and route tables as supporting evidence. This assignment only defines the starting direction for

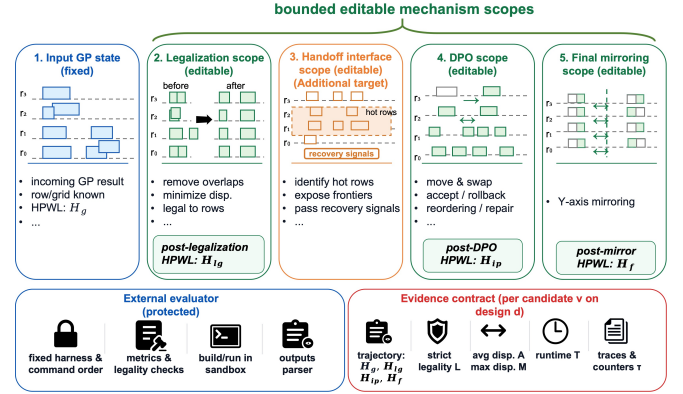


Figure 3: OpenROAD DPL instantiation. ReviewDSE edits bounded legalization, DPO, and handoff mechanisms while keeping the evaluation path protected.

the target search. The selected branch is not fixed across iterations. Later rounds may extend it if it remains effective, or switch to repairs, alternative branches, or new mechanisms when target-local evidence suggests a better route.

In later iterations, the Teacher first reviews the previous round and then decides how to balance continuation and exploration. A clean candidate with the best final objective inside the runtime gate may seed an elite continuation branch, typically for one Student. The other Students can explore diversity routes or reuse evidence from peer candidates. This reuse remains source-level: a Student may inspect another candidate’s diff, summarize its logic chain, adapt a patch, switch to a branch, or reimplement the idea in its own workspace. The reviewed mechanism guides the edit, but it is not copied as an independent plug-in.

Each Student edits a private optimizer source workspace and submits a complete $O + \Delta$ candidate. The Student must build or relink a private binary, run the fixed evaluator with that binary, and provide complete artifacts. Because earlier-stage choices affect later stages, the Student is encouraged to reason about the whole stage trajectory rather than optimizing a stage metric in isolation.

After evaluation, the Teacher reviews the complete-flow outcome together with the mechanism evidence behind it, and assigns a review decision following Table 3. A valid improved candidate may become the promoted parent for later target search. Other evaluated candidates may still be retained as target-local evidence when their source diffs and logs explain an active mechanism, even if they are not promoted. Invalid candidates are recorded but cannot be promoted. These review outcomes update only the target-local record, while the Level 1 warm-start evidence remains unchanged.

3.5 Instantiation on OpenROAD Detailed Placement

We instantiate ReviewDSE on OpenROAD detailed placement.¹ We refer to the source-specialized commands as DPL-evolve, DPO-evolve, and mirror-evolve. Figure 3 summarizes the editable and protected parts of this instantiation. The evaluated command sequence is fixed and keeps the canonical three stages: legalization, DPO, and mirroring. ReviewDSE edits only the bounded source implementations

¹We use constructed Nangate45 detailed-placement calibration cases for Level 1: JPEG at UTIL=90, AES at UTIL=70, and a SWERV wrapper at UTIL=60.

invoked by these stages in private workspaces. The strict comparison anchor is the fixed OpenROAD detailed-placement flow.

Following the formulation above, we use legalization and DPO as the two main stage scopes, and we use the legalization-DPO handoff as the main interface scope. In legalization, source-start branches include Diamond-style, Negotiation-based, and LEGALM-inspired methods. Within this scope, ReviewDSE may explore row-assignment policies, row-segment reservation, overflow-aware relocation, connectivity-aware escape, bounded compaction, and post-legality repair.

DPO exploration starts from the same native improve-placement source inside the selected legalizer branch. The Teacher assigns the mechanism implementation path, and the Student edits the relevant DPO source path. The native improve-placement behavior contains local kernels such as independent-set matching, global swaps, vertical swaps, row-local reordering, randomized local improvement, and legality or orientation checks. ReviewDSE may revise or extend this source with local-move generation, partner selection, region kicks, dirty-window dynamic programming, exact affected-net HPWL scoring, transaction accept/rollback, fallback descent, per-net extrema caches, and thread-local scratch storage. Handoff mechanisms include dirty rows, pressure regions, touched-net frontiers, boundary-active cells, local repair windows, and source-produced state that downstream DPO can consume.

For an evaluated candidate $O+\Delta$ on design d , the evaluator records the full stage trajectory

$$E(O + \Delta, d) = \{H_g, H_{lg}, H_{ip}, H_f, L, A, M, T, \tau\}, \quad (3)$$

where H_g , H_{lg} , H_{ip} , and H_f are half-perimeter wirelength (HPWL) after incoming global placement, legalization, DPO, and final post-DPL evaluation. L denotes strict legality, A and M are average and maximum displacement, T is runtime, and τ contains traces, liveness counters, and diagnostics. Full-trajectory review is necessary because a stage-local improvement is not always a valid target improvement. A lower legalization HPWL H_{lg} may still worsen the final post-DPL HPWL H_f after DPO and mirroring. Conversely, a non-winning candidate can also provide active mechanism evidence under specific target patterns. This is why ReviewDSE retains mechanism evidence for later target search.

4 Experiments

Our experiments compare white-box source optimization with black-box DSE, test runtime-aware selection, examine full-flow composability, and evaluate whether source-mechanism exploration can recover legality on hard cut-row layouts.

4.1 Experiment Setup and Metrics

We use a fixed OpenROAD/OpenROAD-flow-scripts checkout in the protected evaluator and run all methods through the same measurement path. The exact checkout, evaluator scripts, and selected-candidate artifacts are recorded with the experiment data. The main target search uses one Teacher agent with the GPT-5.5 xhigh profile and four parallel Student agents with the GPT-5.4 xhigh profile for 10 review iterations. We report only candidates that compile, pass strict legality, produce complete protected metrics, satisfy

Table 4: Nine-case DSE results.

Design	Default	BO	ReviewDSE-HPWL	ReviewDSE- G_{HR}	Tokens
AES ASAP7	42.2k/5.3s	-0.98%/1.47×	-1.91%/1.91× i10	-1.40%/1.38× i10	1.62/0.08
AES N45	176.8k/6.0s	-1.20%/1.38×	-3.67%/1.86× i10	-3.67%/1.86× i10	1.81/0.09
Ariane133 N45	5.78M/151s	-0.06%/0.92×	-5.50%/0.56× i10	-5.50%/0.56× i10	3.66/0.11
Ibex ASAP7	63.3k/6.9s	-0.10%/1.10×	-0.60%/1.17× i10	-0.60%/1.17× i10	1.68/0.10
Ibex N45	200.1k/5.1s	-0.09%/1.00×	-0.99%/2.00× i10	-0.56%/1.08× i6	1.52/0.12
JPEG ASAP7	134.6k/23s	-0.63%/1.39×	-0.94%/1.00× i5	-0.94%/1.02× i3	1.87/0.10
JPEG N45	467.2k/33s	-0.23%/1.13×	-1.63%/0.56× i9	-1.63%/0.77× i10	1.93/0.09
SWERV ASAP7	913.9k/57s	-0.04%/1.08×	-0.33%/1.90× i8	-0.30%/1.09× i4	3.06/0.09
SWERV N45	3.27M/34s	-0.09%/1.08×	-0.49%/1.07× i9	-0.49%/1.07× i9	2.14/0.14
Mean	–	-0.38%/1.17×	-1.78%/1.34×	-1.68%/1.11×	2.15/0.10

Entries are post-DPL HPWL delta/runtime vs. default; $i\#$ is the selected iteration. Tokens are logged/active budgets in billions.

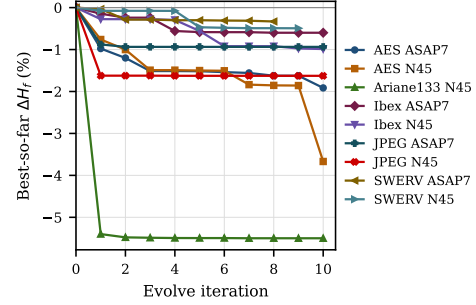


Figure 4: 10-iteration best-so-far ΔH_f .

source/binary/metric consistency, show liveness, and stay within a $2\times$ runtime gate.

We compare against BO-DSE, a representative black-box DSE baseline that keeps the OpenROAD source fixed. BO-DSE uses Bayesian optimization (BO) with Optuna’s Tree-structured Parzen Estimator (TPE) sampler. It searches exposed legalization, DPO, extra-invocation, and global-swap controls for 400 trials per case with four-way parallelism.

Our experimental dataset comprises nine target tasks derived from the AES, Ariane133, Ibex, JPEG, and SWERV designs, implemented in ASAP7 and Nangate45 technology nodes. The calibration and target sets are disjoint evaluation instances. They may share benchmark families, but no target placement, target metric, target route, or target-specific candidate is used in Level 1. All target designs are evaluated independently by the protected flow.

To evaluate quality of results (QoR), we use the final post-DPL HPWL, denoted as H_f , as the primary metric. We report the relative HPWL change ΔH_f and runtime compared with the default Diamond detailed-placement flow. For ReviewDSE, each test case reports both the best- H_f candidate that satisfies the $2\times$ runtime gate and a lower-overhead candidate selected by a runtime-aware score G_{HR}^2 .

4.2 White-Box Source Optimization

Table 4 compares the OpenROAD default, public-knob BO-DSE, and ReviewDSE on the nine target tasks. After 400 trials per case, BO-DSE improves mean post-DPL HPWL by 0.38%, while ReviewDSE improves all nine targets by 1.78% on average within the same $2\times$

²The higher-is-better selection score is defined as $G_{HR} = 100 \cdot (H_f^{Default} - H_f) / H_f^{Default} - P(T/T^{Default})$, where T is the runtime, $P(r) = 0$ for $r \leq 1.10$, and $P(r) = (\sqrt{r} - \sqrt{1.10}) / (\sqrt{2} - \sqrt{1.10})$ for $r > 1.10$. A $2\times$ runtime penalty requires a one percentage point reduction in post-DPL HPWL to break even; small runtime noise is ignored.

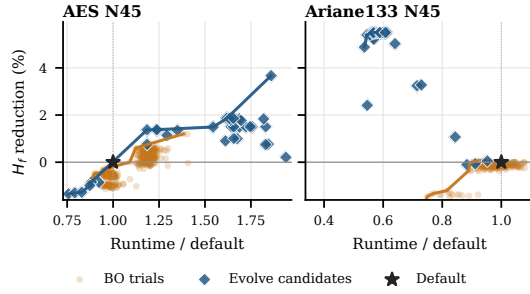


Figure 5: Runtime-quality frontier on two Nangate45 cases; y-axis is post-DPL HPWL reduction vs. default.

runtime gate. Runtime-aware ReviewDSE- G_{HR} preserves a 1.68% mean HPWL reduction while reducing average runtime from 1.34 $\times$ to 1.11 $\times$. This comparison uses the same protected evaluator and target tasks, but the two methods search different spaces: BO-DSE searches exposed controls of a fixed source tree, while ReviewDSE explores reviewed source mechanisms. Calibration evidence only initializes promising source directions. Every reported candidate is rebuilt, evaluated, and re-validated on its own target by the protected evaluator. The token column reports mean per-target search cost.

Figure 4 shows best-so-far ΔH_f over 10 target iterations. Early drops indicate that Level 1 evidence can choose useful initial source directions.³ Later drops, such as AES N45, reflect target-specific source exploration after Teacher review, runtime feedback, and peer-candidate evidence. Flat segments correspond to iterations without a promoted parent, often because review rejects non-composable or runtime-expensive candidates.

Figure 5 plots HPWL reduction against normalized runtime for two representative Nangate45 cases. On AES N45, ReviewDSE reaches a 3.67% HPWL reduction and outperforms the best BO-DSE point at comparable runtime. On Ariane133 N45, ReviewDSE reaches a 5.50% HPWL reduction while reducing runtime relative to the default. These tradeoffs were not observed in the 400-trial BO-DSE search over the specified public-knob space.

4.3 Full-Flow Review and Downstream Composability

We next examine why ReviewDSE reviews complete trajectories instead of accepting stage-local winners. The test applies a deliberately stage-local selection rule: it selects candidate legalizers by post-legalization HPWL H_{lg} , then passes the selected placements through the same downstream DPO and final mirroring flow. The dense N45 cases are constructed validation variants with increased legalization pressure and are not included in the nine-case QoR average. Table 5 reports cases where the selected legalizer improves H_{lg} but degrades the final post-DPL HPWL H_f .

These counterexamples show that a lower-HPWL legalized placement can be less compatible with downstream DPO. JPEG dense N45 is the clearest case: the selected legalizer reduces H_{lg} by 14.96%,

³ In our tested Ariane133 N45 diagnostic runs, missing Diamond-sourceTopK averaged +1.52%/1.07 $\times$. First-iteration Level 1 guidance matched the case feature, selected it, and averaged -3.26%/0.71 $\times$, giving later rounds a local parent instead of spending early target iterations only on route discovery. Diamond-sourceTopK scores top-K DPO moves/swaps from a Diamond basin and passes hot nodes and segments to reorder. This diagnostic is not used as a separate evaluation target.

Table 5: Stage-local legalization counterexamples.

Case	Selected legalizer	Full-flow reference	H_{lg} / ref.	ΔH_{lg}	H_f / ref.	ΔH_f
AES dense N45	LEGALM	Diamond	137.8k / 138.8k	-0.76%	130.8k / 128.9k	+1.48%
JPEG dense N45	Negotiation	Negotiation	1.26M / 1.48M	-14.96%	711.7k / 588.3k	+20.96%
SWERV dense N45	Diamond	Negotiation	2.81M / 2.81M	-0.12%	2.71M / 2.71M	+0.02%

Table 6: Pattern-level hard cut-row repair evidence.

Case	Cut-Row Pattern	Fixed ref (D/N)	ReviewDSE	Evidence
Ariane133 N45	center-8um	fail/pass	pass, 42.2s	beats legal N by 1.03%, 18.9 $\times$
	center-9um	fail/fail	pass, 51.6s	strict repair; no fixed legal ref.
	center-10um	fail/TO	pass, 59.6s	strict repair; N reaches timeout
SWERV dense N45	center-5.25um	pass/fail	pass, 48.6s	beats legal fixed ref. by 3.69%, 3.4 $\times$
	center-5.5um	fail/fail	pass, 50.8s	strict repair; no fixed legal ref.
	center-6um	fail/fail	pass, 51.7s	strict repair; no fixed legal ref.
BPQUAD	center-5um	fail/TO	pass, 563.6s	D fails; N reaches 7200s cap
	center-6um	fail/TO	pass, 799.4s	D fails; N reaches 7200s cap
	center-8um	fail/TO	pass, 1085.0s	D fails; N reaches 7200s cap

D/N = Diamond/Negotiation; TO = 7200s timeout. center-xum is a vertical center-band cut-row with $x\mu m$ halo. pass means OpenROAD and check_placement pass.

but the final H_f worsens by 20.96% after DPO and mirroring. Thus, legalization quality alone is not a sufficient promotion criterion.

4.4 Hard Cut-Row Repair

Finally, we test legality recovery on hard cut-row stress tests. These layouts contain fragmented row topologies that cause fixed-source reference routes to fail legality, time out, or miss a strict legal solution. Unlike the nine-case QoR comparison, the objective here is to recover legal placements under the same protected legality checker and runtime cap.

Table 6 reports pattern-level replay evidence rather than only family-level counts. ReviewDSE produces strict legal placements on all nine hard cut-row patterns. For Ariane133 and SWERV, at least one pattern also has a legal fixed-source reference, and ReviewDSE improves both HPWL and runtime. For BPQUAD, the fixed routes fail or time out, so the evidence is legality recovery within 563.6–1085.0s rather than QoR dominance. These results complement Table 4 by showing that reviewed source-mechanism exploration can recover strict legality on failure-oriented stress tests, not only improve QoR on already feasible targets.

5 Discussion

ReviewDSE intentionally performs target-case source specialization, not universal optimizer replacement. Calibration provides method evidence and source starts, but each reported candidate is rebuilt, executed, and promoted only on its own target under the protected evaluator. The fixed measurement path prevents metric gaming, and promotion also requires canonical metrics, source/binary/metric consistency, runtime gating, and liveness evidence.

The main limitations are evaluator scope and search cost. Current results cover HPWL, runtime, displacement, and strict legality; timing, congestion, power, and signoff require additional protected metrics. The target searches average 2.15B logged tokens and 0.10B active uncached-input-plus-output tokens over 10 iterations, so ReviewDSE is most appropriate for high-value designs, hard failures, or cases where public knobs have limited remaining headroom.

6 Conclusion

ReviewDSE reframes open-source EDA tuning as protected white-box DSE: for a target design, bounded optimizer mechanisms can be

specialized under a fixed evaluator rather than only invoked through public knobs. In OpenROAD detailed placement, this source-level search boundary improves the post-DPL HPWL/runtime frontier beyond public-knob BO-DSE, reveals why stage-local winners require full-flow review, and recovers legality on hard cut-row stress tests. The main lesson is that source-mechanism exploration for EDA should not be treated as unrestricted tool rewriting: complete candidates must be judged by canonical metrics and legality checks, while reusable knowledge is accumulated as reviewed mechanism evidence.

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A Artifact Appendix

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A.1 Abstract

The artifact contains ReviewDSE, its prompts and protected evaluator, a pinned open-source EDA setup, retained source programs, and scripts for Tables 4–6 and Figures 4–5. It was tested on Rocky Linux 8.10/x86-64; no GPU, commercial EDA license, or commercial PDK is required. The LLM Agent requires an authenticated Codex installation. The evaluated snapshot is archived at [ZENODO DOI AND URL], with the live repository at <https://github.com/yuan-fd/DPLEvolve-AE>.

A.2 Artifact Check-list (Meta-information)

- **Algorithm/program:** Protected Teacher–Student white-box DSE of OpenROAD detailed placement on nine targets and nine cut-row patterns; Agents edit bounded sources and full-flow review selects candidates.
- **Tools/build/binary:** Pinned OpenROAD, ORFS, Yosys, OpenSTA, GCC/G++ 9+, and CMake 3.16+. Tools are built from source; each candidate uses a private OpenROAD binary.
- **Models/access/prompts:** Paper: one gpt-5.5 Teacher and four gpt-5.4 Students. For AE, gpt-5.6-sol is the Teacher and gpt-5.6-terra is the Student; all use xhigh. Hosted access through authenticated Codex CLI is required. Templates, prompt construction, and prompt records are included; other decoding settings use service defaults.
- **Data:** Table 4 inputs are regenerated by ORFS; Table 5 includes three checksummed source snapshots; Table 6 uses a separately downloaded ~200-MB DEF/Verilog/SDC archive.
- **Environment/hardware:** Linux x86-64, Bash 4+, Make 4+, Python 3.11+, and OpenROAD dependencies. Reference: Rocky Linux 8.10, 2× Xeon Platinum 8462Y+, 64 physical/128 logical cores, 314 GiB RAM; CPU-only.
- **Execution/resources:** At least 100 GiB free disk is recommended. Setup takes several hours; full BO and model-search campaigns take multiple days. Parallelism can be reduced.
- **Metrics/output:** Stage HPWL, strict legality, displacement, runtime, liveness, and source/binary/metric consistency; fresh JSON/TSV, logs, provenance, review records, and SVGs.
- **Experiments/results:** Tables 4–6, Figures 4–5, selected-source replay, Level 1 calibration, and Level 2 search; paper values and tolerances are checked in Section A.6.
- **Cost:** Mean paper usage is 2.15B logged tokens (~0.10B active tokens) per target; monetary cost depends on the account and dated provider prices.
- **Availability/licenses:** Public BSD-3-Clause code; no proprietary component. Third-party tools, data, and hosted models retain their licenses and terms.
- **Workflow/archive:** GNU Make, Bash, Python, ORFS, and Codex CLI; Zenodo: [ZENODO DOI AND URL].

A.3 Description

Access and dependencies. The Zenodo snapshot and live repository above contain identical entry points and integrity metadata;

detailed commands are in the README and docs/reviewer-walkthrough.md. Setup needs Linux x86-64, Git, rsync, the build packages listed above, and network access, but no root or commercial software. The fixed OpenROAD source is upstream branch master, base commit d14d526a6f8c; the prepared artifact commit is d5ff63abel1a1. Full SHA-1 values and pinned ORFS, Yosys, and yosys-slang revisions are in provenance/source-commits.json.

Data and models. Only Table 5 input preparation sets the ORFS CORE_UTILIZATION variable to 70/90/60 for AES/JPEG/SWERV; it does not alter the global ORFS configs or Table 4 cases. Its six roles reuse the archived LEGALM/Diamond/Negotiation snapshots: AES uses LEGALM/Diamond, JPEG uses Negotiation/Negotiation, and SWERV uses Diamond/Negotiation. Diamond and Negotiation are OpenROAD implementations; the Agent implemented LEGALM from the published method. Table 6 contains exact DEF, Verilog, and SDC inputs. The paper search ran in April–May 2026 with the models listed above. Codex does not expose a stable service revision; the harness leaves temperature, top-p, top-k, seed, stops, and a separate token limit at service defaults, and retries up to eight times with 45-s linear backoff. Prompts are under src/dpl_evolve_agent/prompt_templates and src/dpl_evolve_agent/scripts/teacher_loop; proposals are stochastic.

A.4 Algorithm Flows

Algorithm 1 freezes calibration evidence and source starts; Algorithm 2 uses them in a target-local Teacher–Student loop. Both call the protected evaluator, and Level 2 never changes the packet.

Algorithm 1 Level 1 calibration and warm-start construction.

Input: Calibration cases $\mathcal{C}$, source O , evaluator E , source starts $\mathcal{B}_0, S_C$ Students
Output: Frozen packet $W = (K, \mathcal{B})$

```

1:  $K \leftarrow \emptyset; \mathcal{B} \leftarrow \mathcal{B}_0$ 
2: for all  $c \in \mathcal{C}$  do
3:    $b_c \leftarrow \text{Evaluate}(E, O, c); Q_c \leftarrow \text{TeacherPlan}(c, b_c, K, \mathcal{B})$ 
4:   for all  $s \in \{1, \dots, S_C\}$  in parallel do
5:      $O_s \leftarrow \text{Instantiate}(\mathcal{B}, Q_c[s])$ 
6:      $\Delta_s \leftarrow \text{StudentEdit}(O_s, Q_c[s])$ 
7:      $\hat{O}_s \leftarrow \text{Build}(O_s + \Delta_s)$ 
8:      $a_s \leftarrow \text{Evaluate}(E, \hat{O}_s, c)$ 
9:   end for
10:   $(K, \mathcal{B}) \leftarrow \text{TeacherReview}(\{a_s\}, K, \mathcal{B})$ 
11: end for
12: return  $W \leftarrow \text{Freeze}(K, \mathcal{B})$ 
```

Each Student returns a complete source reference, diff, full-flow metrics, legality, runtime, and liveness. Teacher review freezes useful evidence and complete starts before target optimization; calibration uses AES/JPEG/SWERV at 70/90/60 utilization.

The frozen packet W contains calibration-reviewed mechanism evidence and complete source starts; it remains read-only during target search. For each target, Students edit private workspaces and return source references, diffs, protected-evaluator outputs, and liveness evidence. Promotion requires build, legality, canonical-metric, source/binary-consistency, liveness, and runtime-gate checks. Teacher review retains other informative outcomes in target-local L_d for later iterations without changing W .

Algorithm 2 Level 2 Teacher–Student target search.

Input: Target d , source O , evaluator E , frozen packet W , S Students, R iterations, runtime gate g

Output: Promoted candidate p_d and target-local record L_d

```

1:  $b_d \leftarrow \text{Evaluate}(E, O, d)$ ;  $L_d \leftarrow \{b_d\}$ ;  $p_d \leftarrow O$ 
2:  $\text{CreatePrivateWorkspaces}(S, O)$ 
3: for  $r = 1$  to  $R$  do
4:    $Q_r \leftarrow \text{TeacherPlan}(d, b_d, W, L_d, p_d)$ 
5:   for all  $s \in \{1, \dots, S\}$  in parallel do
6:      $O_s \leftarrow \text{Instantiate}(p_d, W, Q_r[s])$ 
7:      $\Delta_s \leftarrow \text{StudentEdit}(O_s, Q_r[s])$ 
8:      $\bar{O}_s \leftarrow \text{Build}(O_s + \Delta_s)$ 
9:      $a_s \leftarrow \text{Evaluate}(E, \bar{O}_s, d)$ 
10:  end for
11:   $V_r \leftarrow \text{ContractValid}(\{a_s\}, g)$ 
12:   $L_d \leftarrow L_d \cup \text{TeacherReview}(\{a_s\}, V_r)$ 
13:   $p_d \leftarrow \text{PromoteBest}(V_r, p_d)$ ;  $\text{PreparePeerEvidence}(L_d)$ 
14: end for
15: return  $(p_d, L_d)$ 

```

▷ W remains frozen

A.5 Installation and Workflow

From the artifact root, prepare the pinned environment and run the smoke and mandatory model-access checks:

```

make doctor
make bootstrap
make build-tools THREADS=8
make check
make prepare-paper-inputs CASE=aes_nangate45 THREADS=8
make validate-evaluator CASE=aes_nangate45 THREADS=8
make check-demo-models
make check-demo-models sends small requests and must re-
port MODEL_READY for both Teacher and Student. These are two
model-readiness requests; a live closed-loop search is not required
for the functional check. Reported experiments use:

bash artifacts/01-table4-qor/reproduce.sh \
  --threads 10
make check-table5-data
make reproduce-table5 THREADS=10
bash artifacts/03-table6-cutrow/reproduce.sh \
  --fetch --threads 10
bash artifacts/04-figures/reproduce.sh
make reproduce-paper-search \
  ACKNOWLEDGE_LLM_COST=yes THREADS=10

```

The first five commands validate the reported results using nine defaults, 400 BO trials per target, 18 selected-program replays, six Table 5 role runs, 27 cut-row runs, and figure regeneration. They retain the mandatory Agent configuration but do not replace the model-backed method. The final command reruns the complete stochastic search and is shown for completeness; it is cost-gated and is not required for the functional check.

A.6 Evaluation and Expected Results

Scripts derive fresh metrics before comparison; expected files never populate observed fields.

- **Table 4 (full rerun):** nine cases produce `table4/table4-fresh.tsv`. Mean HPWL changes are -0.38% (BO), -1.78% (HPWL), and -1.68% (GHR); the latter runtimes are $1.34 \times / 1.11 \times$.

All runs must be legal and fall within 0.06 percentage point and 0.20 runtime-ratio windows.

- **Table 5 (full rerun):** six rebuilt role runs produce `table5_*/table5-fresh.tsv`. Each pair must satisfy $\Delta H_{I_g} < 0$ and $\Delta H_f > 0$; AES, JPEG, and SWERV report $-0.76\%/+1.48\%$, $-14.96\%/+20.96\%$, and $-0.12\%/+0.02\%$.
- **Table 6 (full rerun):** `table6_*/table6-fresh.tsv` has 27 rows. Status and strict-legality classifications must match exactly, with a 35% runtime window: nine ReviewDSE, one Diamond, and one Negotiation cases are legal.
- **Figures 4–5 (retained-output validation):** Figure 4 contains 96 observed points and lists three absent SWERV points without imputation; Figure 5 recomputes runtime-ratio Pareto membership. TSV/SVG outputs are under `paper_reproduction/figures/`.

Level 1 is a runnable reconstruction because the author-time Student breadth and frozen packet were not retained. A stochastic new search is checked by its protected evaluation and fresh trajectory, not identical proposals.

A.7 Integrity and Failure Reporting

Before reporting results, verify the repository and release archive:

```

make test
make validate-configs
make zenodo-audit

```

The archive audit enumerates the three Table 5 program snapshots and checks their SHA-256 manifest. Table 6 is accepted only after its downloaded manifest passes. Fresh outputs are written below `DPL_EVOLVE_STATE_ROOT`; expected files and retained plots are read-only comparison inputs. A failed build, legality check, metric contract, or numerical window returns nonzero and keeps the command, manifest, result directory, and first failure for review.

A.8 Experiment Customization

CASE, THREADS, STUDENTS, and ITERATIONS select the target and concurrency/search size; Teacher/Student model variables are exposed. Figure inputs may be retained or fresh, and a new campaign uses a distinct `DSE_RUN_PREFIX`. Candidate selection keeps the paper’s exact $2 \times$ runtime gate; paper-scale model execution additionally requires the explicit cost acknowledgement shown above.

A.9 Methodology

Submission, reviewing, and badging methodology:

- <https://www.acm.org/publications/policies/artifact-review-and-badging-current>
- <http://cTuning.org/ae/submission-20201122.html>
- <https://github.com/ml-eda/artifact-evaluation/>